

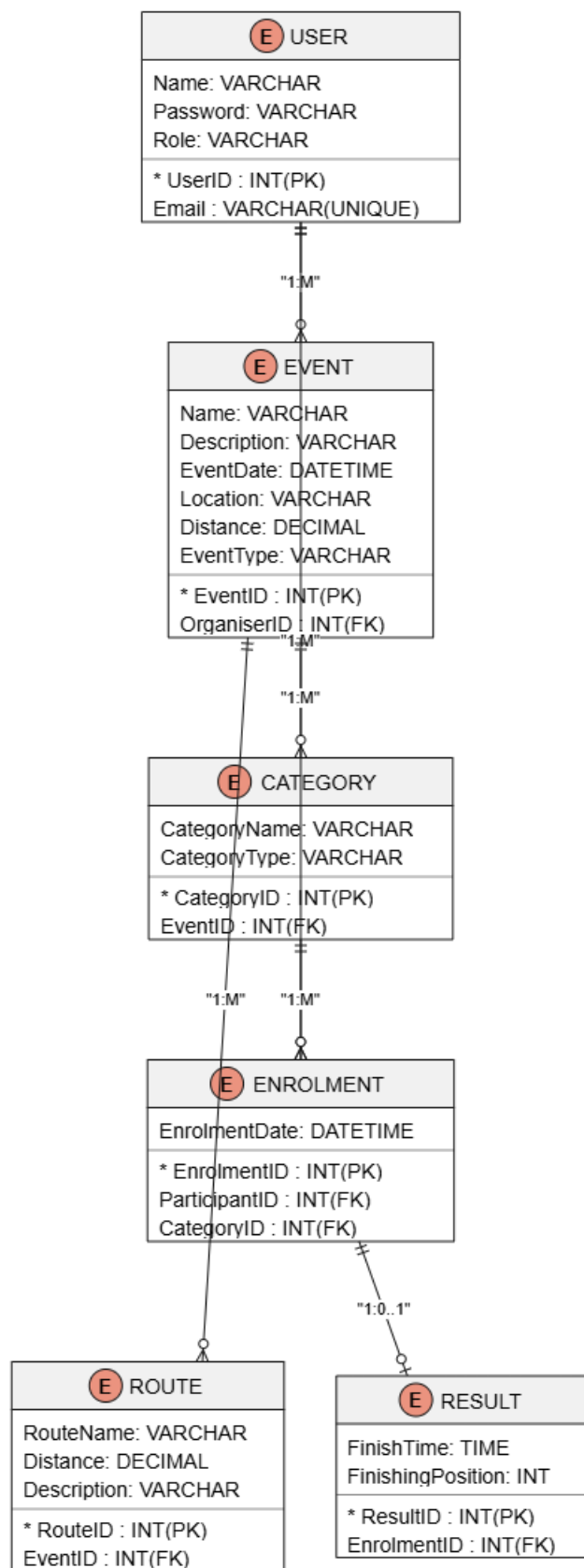
SESI MONAGATANE

ST-10468899

PROGRAMMING 6212

PART 1

SECTION A



SECTION B

The RaceDay API endpoint plan defines the HTTP methods, routes, access roles, request information and expected responses required to support the functional requirements of the system. The endpoints provide functionality for authentication, user profiles, event management, category management, event enrolments and participant results. Role-based access is used to ensure that Organisers and Participants can only perform actions appropriate to their roles.

Authentication

HTTP method	Route	Description	Role required	Request body	Expected Response
POST	/api/auth/register	Registers a new RaceDay user as a Participant or Organiser.	None	Name, Email, Password, Role	201-User created successfully
POST	/api/auth/login	Authenticates a user and provides access according to their role.	None	Email, password	400-Login successful 401-Unauthorized-Invalid credentials

User Profile

HTTP method	Route	Description	Role required	Request body	Expected response
GET	/api/users/profile	Allows a logged-in user to view their own profile.	Participant / Organiser	None	200 -Profile details returned

POST	/api/users/profile	Allows a logged-in user to update their own profile.	Participant / Organiser	Name, Email	200-Profile updated successfully
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EVENTS

HTTP method	Route	Description	Role required	Request body	Expected response
GET	/api/events	Retrieves a list of upcoming events	Participant /organiser	None	200-List of events returned
GET	/api/events/{id}	Retrieves details of a specific event	Participant/organiser	none	200-Event details returned
POST	/api/events	Creates a new event	organiser	Name, Description, Date, Location, Distance, EventType	201-Event created successfully
PUT	/api/events/{id}	Updates an existing event	organiser	Name, Description, Date, Location, Distance, EventType	200-Event updated successfully
DELETE	/api/events/{id}	Deletes an existing event	organiser	None	200-Event deleted successfully

EVENT ENROLMENTS

HTTP method	Route	description	Role required	Request body	Expected response
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POST	/api/events/{eventId}/enrolments	Allows a participant to enter an event and select a category	Participant	category	201-Enrolment created successfully
GET	/api/events/{eventId}/enrolments	Allows an organiser to view participants enrolled in their event	organiser	None	200-List of enrolments returned
GET	/api/users/profile/enrolments	Allows a participant to view their own event enrolments	participant	None	200-Participant's enrolments returned

RESULTS

HTTP method	Route	Description	Role required	Request body	Expected response
POST	/api/events/{eventId}/results	Records a participant's finish time and finishing position	organiser	ParticipantID, FinishTime, FinishingPosition	201-Result recorded successfully
GET	/api/events/{eventId}/results	Allows an organiser to view results for their event.	organiser	None	200-Event results returned
GET	/api/users/profile/results	Allows a participant to view their own results and	Participant	none	200-Participant results returned

		performan ce history.			
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SECTION C

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1  -- Step 1: Create and use the RaceDay Database
2  CREATE DATABASE RaceDayDB;
3  GO
4
5
6
7  -- 2. Create USER Table
8  CREATE TABLE [USER] (
9      UserID INT IDENTITY(1,1) PRIMARY KEY,
10     Name VARCHAR(100) NOT NULL,
11     Email VARCHAR(150) NOT NULL UNIQUE,
12     Password VARCHAR(255) NOT NULL,
13     Role VARCHAR(50) NOT NULL
14 );
15 GO
16
17 -- 3. Create EVENT Table
18 CREATE TABLE EVENT (
19     EventID INT IDENTITY(1,1) PRIMARY KEY,
20     OrganiserID INT NOT NULL,
21     Name VARCHAR(100) NOT NULL,
22     Description VARCHAR(MAX) NULL,
23     EventDate DATETIME NOT NULL,
24     Location VARCHAR(100) NOT NULL,
25     Distance DECIMAL(5,2) NOT NULL,
26     EventType VARCHAR(50) NOT NULL,
27     CONSTRAINT FK_Event_User FOREIGN KEY (OrganiserID) REFERENCES [USER](UserID) ON DELETE CASCADE
28 );
29 GO
30
31 -- 4. Create CATEGORY Table
32 CREATE TABLE CATEGORY (
33     CategoryID INT IDENTITY(1,1) PRIMARY KEY,
34     EventID INT NOT NULL,
35     CategoryName VARCHAR(100) NOT NULL,
36     CategoryType VARCHAR(50) NOT NULL,
37     Description VARCHAR(MAX) NULL,
38     CONSTRAINT FK_Category_Event FOREIGN KEY (EventID) REFERENCES EVENT(EventID) ON DELETE CASCADE
39 );
40 GO
41
42 -- 5. Create ROUTE Table
43 CREATE TABLE ROUTE (
44     RouteID INT IDENTITY(1,1) PRIMARY KEY,
45     EventID INT NOT NULL,
46     RouteName VARCHAR(100) NOT NULL,
47     Distance DECIMAL(5,2) NOT NULL,
48     Description VARCHAR(MAX) NULL,
49     CONSTRAINT FK_Route_Event FOREIGN KEY (EventID) REFERENCES EVENT(EventID) ON DELETE CASCADE
50 );
51 GO
52
53 -- 6. Create ENROLMENT Table
54 CREATE TABLE ENROLMENT (
55     EnrolmentID INT IDENTITY(1,1) PRIMARY KEY,
56     ParticipantID INT NOT NULL,
57     CategoryID INT NOT NULL,
58     EnrolmentDate DATETIME DEFAULT GETDATE(),
59     CONSTRAINT FK_Enrolment_Participant FOREIGN KEY (ParticipantID) REFERENCES [USER](UserID),
60     CONSTRAINT FK_Enrolment_Category FOREIGN KEY (CategoryID) REFERENCES CATEGORY(CategoryID)
61 );
62 GO
63
64 -- 7. Create RESULT Table
65 CREATE TABLE RESULT (
66     ResultID INT IDENTITY(1,1) PRIMARY KEY,
67     EnrolmentID INT NOT NULL UNIQUE,
68     FinishTime TIME NOT NULL,
69     FinishingPosition INT NOT NULL,
70     CONSTRAINT FK_Result_Enrolment FOREIGN KEY (EnrolmentID) REFERENCES ENROLMENT(EnrolmentID) ON DELETE CASCADE
71 );
72 GO
```

Results

Messages

	UserID	Name	Email	Password	Role	
1	1	Alice Organiser	alice@raceday.com	hashed_pwd_123	Organiser	
2	2	Bob Runner	bob@raceday.com	hashed_pwd_456	Participant	

	EventID	OrganiserID	Name	Description	EventDate	Location	Distance	EventType	
1	1	1	City Marathon 2026	Annual city marathon event	2026-11-15 06:00:00.000	Central Park	42.20	Run	
2	2	1	City Marathon 2026	Annual city marathon event	2026-11-15 06:00:00.000	Central Park	42.20	Run	
3	3	1	City Marathon 2026	Annual city marathon event	2026-11-15 06:00:00.000	Central Park	42.20	Run	
4	4	1	City Marathon 2026	Annual city marathon event	2026-11-15 06:00:00.000	Central Park	42.20	Run	
5	5	1	City Marathon 2026	Annual city marathon event	2026-11-15 06:00:00.000	Central Park	42.20	Run	

	ResultID	EnrolmentID	FinishTime	FinishingPosition	
1	1	1	00:45:30.0000000	1	